



NCI Protocol #: 10323

Patient ID:

Date:

1 of 12

Patient Name:

Date Collected:

Biopsy Site: **Right medial gluteal muscle**

Sample ID:

Telephone:

Tumor Content (%): **50_69%**

Patient DOB:

Referring Physician:

Primary Diagnosis: **Metastatic melanoma**

MoCha ID:

Fax:

Preanalytics to include tumor enrichment as required is performed by Van Andel Research Institute; Grand Rapids; MI 49503

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Relevant Melanoma Findings

Gene	Finding
BRAF	BRAF V600K
KIT	None detected
NRAS	None detected
NTRK1	None detected
NTRK2	None detected
NTRK3	None detected

Relevant Biomarkers

Tier	Genomic Alteration	Relevant Therapies (In this cancer type)	Relevant Therapies (In other cancer type)	Clinical Trials
IA	BRAF V600K	atezolizumab + cobimetinib + vemurafenib ¹ binimetinib + encorafenib ¹ cobimetinib + vemurafenib ¹ dabrafenib + trametinib ¹ trametinib ¹ dabrafenib encorafenib vemurafenib	dabrafenib vemurafenib	41
III	RAD51D R188Q, RAD51D R53W	None	niraparib ¹	7

Public data sources included in relevant therapies: FDA¹, NCCN

Tier Reference: Li et al. Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer: A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists. J Mol Diagn. 2017 Jan;19(1):4-23.

Variant Details

DNA Sequence Variants

Gene	Amino Acid Change	Coding	Variant ID	Locus	Allele Frequency	Transcript	Variant Effect
BRAF	p.(V600K)	c.1798_1799delGTins AA	COSM473	chr7:140453136	74.56%	NM_004333.4	missense
RAD51D	p.(R188Q)	c.563G>A	.	chr17:33428224	38.17%	NM_133629.2	missense
RAD51D	p.(R53W)	c.157C>T	.	chr17:33433488	33.47%	NM_133629.2	missense

Comments:

A splice variant in CDKN2A gene, c.151-2A>T, is also identified at ~66% VAF in this patient's specimen. This variant has been previously reported in multiple cancer types, including malignant melanoma (COSMIC database), and is predicted to result in premature truncation and subsequent loss of CDKN2A protein function. There are currently no FDA approved therapies targeting mutations in CDKN2A gene.

Report Signed By

Name	Role	Date

Biomarker Descriptions

BRAF (B-Raf proto-oncogene, serine/threonine kinase)

Background: The BRAF gene encodes the B-Raf proto-oncogene serine/threonine kinase, a member of the RAF family of serine/threonine protein kinases which also includes ARAF and RAF1 (CRAF). BRAF is among the most commonly mutated kinases in cancer. Activation of the MAPK pathway occurs through BRAF mutations and leads to an increase in cell division, dedifferentiation, and survival^{1,2}. BRAF mutations are categorized into three distinct functional classes namely, class 1, 2, and 3, and are defined by the dependency on the RAS pathway. Class 1 and 2 BRAF mutants are RAS-independent in that they signal as active monomers (Class 1) or dimers (Class 2) and become uncoupled from RAS GTPase signaling, resulting in constitutive activation of BRAF³. Class 3 mutants are RAS dependent as the kinase domain function is impaired or dead^{3,4,5}.

Alterations and prevalence: Recurrent somatic mutations in BRAF are observed in 40-60% of melanoma and thyroid cancer, approximately 10% of colorectal cancer, and about 2% of non-small cell lung cancer (NSCLC)^{6,7,8,9,10}. Mutations at V600 belong to class 1 and include V600E, the most recurrent somatic BRAF mutation across diverse cancer types^{4,11}. Class 2 mutations include K601E/N/T, L597Q/V, G469A/V/R/, G464V/E/, and BRAF fusions⁴. Class 3 mutations include D287H, V459L, G466V/E/A, S467L, G469E, and N581S/I⁴. BRAF V600E is universally present in hairy cell leukemia, mature B-cell cancer, and prevalent in histiocytic neoplasms^{12,13,14}. Other recurrent BRAF somatic mutations cluster in the glycine-rich phosphate-binding loop at codons 464-469 in exon 11 as well as additional codons flanking V600 in the activation loop¹¹. In primary cancers, BRAF amplification is observed in 8% of ovarian cancer and about 1% of breast cancer^{7,10}. BRAF fusions are mutually exclusive to BRAF V600 mutations and have been described in melanoma, thyroid cancer, pilocytic astrocytoma, NSCLC, and several other cancer types^{15,16,17,18,19}. Part of the oncogenic mechanism of BRAF gene fusions is the removal of the N-terminal auto-inhibitory domain leading to constitutive kinase activation^{5,15,17}.

Potential relevance: Vemurafenib²⁰ (2011) was the first targeted therapy approved for the treatment of patients with unresectable or metastatic melanoma with a BRAF V600E mutation. BRAF class 1 mutations, including V600E, are sensitive to vemurafenib, whereas class 2 and 3 mutations are insensitive⁴. BRAF kinase inhibitors including dabrafenib²¹ (2013) and encorafenib²² (2018) are also approved for the treatment of patients with unresectable or metastatic melanoma with BRAF V600E/K mutations. Encorafenib²² is

Biomarker Descriptions (continued)

approved in combination with cetuximab²³ (2020) for the treatment of BRAF V600E mutated colorectal cancer. Due to the tight coupling of RAF and MEK signaling, several MEK inhibitors have been approved for patients harboring BRAF alterations⁴. Trametinib²⁴ (2013) and binimetinib²⁵ (2018) were approved for the treatment of metastatic melanoma with BRAF V600E/K mutations. Combination therapies of BRAF plus MEK inhibitors have been approved in melanoma and NSCLC. The combinations of dabrafenib/trametinib (2015) and vemurafenib/cobimetinib²⁶ (2015) were approved for the treatment of patients with unresectable or metastatic melanoma with a BRAF V600E/K mutation. Subsequently, the combination of dabrafenib and trametinib was approved for metastatic NSCLC (2017) with a BRAF V600E mutation. The PD-L1 antibody, atezolizumab²⁷, has also been approved in combination with cobimetinib and vemurafenib for BRAF V600 mutation-positive unresectable or metastatic melanoma. In 2018, binimetinib²⁸ was also granted breakthrough designation in combination with cetuximab and encorafenib for BRAF V600E mutant metastatic colorectal cancer. The pan-RAF kinase inhibitor, tovorafenib (DAY-101), was granted breakthrough therapy designation (2020) by the FDA for pediatric patients with advanced low-grade glioma harboring activating RAF alterations²⁹. The ERK inhibitor ulixertinib³⁰ was also granted a fast track designation in 2020 for the treatment of patients with non-colorectal solid tumors harboring BRAF mutations G469A/V, L485W, or L597Q. BRAF fusion is a suggested mechanism of resistance to BRAF targeted therapy in melanoma³¹. Additional mechanisms of resistance to BRAF targeted therapy include BRAF amplification and alternative splice transcripts as well as activation of PI3K signaling and activating mutations in KRAS, NRAS, and MAP2K1/2 (MEK1/2)^{32,33,34,35,36,37,38}. Clinical responses to sorafenib and trametinib in limited case studies of patients with BRAF fusions have been reported¹⁹.

RAD51D (RAD51 paralog D)

Background: The RAD51D gene encodes the RAD51 paralog D protein, a member of the RAD51 recombinase family that also includes RAD51, RAD51B (RAD51L1), RAD51C (RAD51L2), XRCC2, and XRCC3 paralogs. The RAD51 family proteins are involved in homologous recombination repair (HRR) and DNA repair of double-strand breaks (DSB)³⁹. RAD51D associates with other RAD51 paralogs to form RAD51B-RAD51C-RAD51D-XRCC2 (BCDX2) complex⁴⁰. The BCDX2 complex binds single- and double-stranded DNA to hydrolyze ATP⁴¹. RAD51D is a tumor suppressor gene. Loss of function mutations in RAD51D are implicated in the BRCAness phenotype, which is characterized by a defect in HRR, mimicking BRCA1 or BRCA2 loss^{42,43}. Germline point mutations in RAD51D are implicated in non-BRCA2 associated breast, ovarian, and colorectal cancer⁴⁴.

Alterations and prevalence: Somatic mutations in RAD51D are rare but have been reported in 1-2% of uterine cancer¹⁰.

Potential relevance: The PARP inhibitor, olaparib⁴⁵ is approved (2020) for metastatic castration-resistant prostate cancer (mCRPC) with deleterious or suspected deleterious, germline or somatic mutations in HRR genes that includes RAD51D. Additionally, consistent with other genes associated with the BRCAness phenotype, RAD51D mutations may aid in selecting patients likely to respond to PARP inhibitors⁴². In 2022, the FDA granted fast track designation to the small molecule inhibitor, pidnarulex⁴⁶, for BRCA1/2, PALB2, or other homologous recombination deficiency (HRD) mutations in breast and ovarian cancers.

Relevant Therapy Summary

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

BRAF V600K

Relevant Therapy	FDA	NCCN	Clinical Trials*
atezolizumab + cobimetinib + vemurafenib	●	●	● (II)
cobimetinib + vemurafenib	●	●	● (II)
binimetinib + encorafenib	●	●	×
dabrafenib + trametinib	●	●	×

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Relevant Therapy Summary (continued)

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

BRAF V600K (continued)

Relevant Therapy	FDA	NCCN	Clinical Trials*
trametinib	●	×	×
vemurafenib	×	◐	● (II)
dabrafenib	×	◐	×
encorafenib	×	●	×
encorafenib, binimetinib	×	×	● (III)
encorafenib, binimetinib, pembrolizumab	×	×	● (III)
nivolumab	×	×	● (II/III)
atezolizumab, tiragolumab	×	×	● (II)
binimetinib, encorafenib	×	×	● (II)
dabrafenib, trametinib	×	×	● (II)
encorafenib, binimetinib, nivolumab, ipilimumab	×	×	● (II)
HL-085, vemurafenib	×	×	● (II)
ipilimumab, binimetinib, nivolumab, encorafenib	×	×	● (II)
RX208	×	×	● (II)
tocilizumab, ipilimumab, nivolumab	×	×	● (II)
trametinib, dabrafenib	×	×	● (II)
vemurafenib, cobimetinib	×	×	● (II)
vemurafenib, cobimetinib, atezolizumab	×	×	● (II)
vemurafenib, cobimetinib, ipilimumab, nivolumab	×	×	● (II)
encorafenib, binimetinib, palbociclib	×	×	● (I/II)
nivolumab, tumor infiltrating lymphocytes, aldesleukin	×	×	● (I/II)
OKI-179, binimetinib	×	×	● (I/II)
RX208, trametinib	×	×	● (I/II)
tazemetostat, trametinib, dabrafenib	×	×	● (I/II)
ABM-1310	×	×	● (I)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Relevant Therapy Summary (continued)

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

BRAF V600K (continued)

Relevant Therapy	FDA	NCCN	Clinical Trials*
BGB-3245	×	×	● (I)
binimetinib	×	×	● (I)
binimetinib, nivolumab, encorafenib	×	×	● (I)
CART-GD2, vemurafenib	×	×	● (I)
cobimetinib, belvarafenib	×	×	● (I)
JSI-1187, dabrafenib	×	×	● (I)
nilotinib, dabrafenib, trametinib	×	×	● (I)
PF 07284890, binimetinib	×	×	● (I)
PF-07799933, cetuximab, binimetinib	×	×	● (I)
RO-7276389, cobimetinib	×	×	● (I)
ZEN-3694, binimetinib	×	×	● (I)

RAD51D R188Q, RAD51D R53W

Relevant Therapy	FDA	NCCN	Clinical Trials*
niraparib	○	×	● (II)
olaparib	×	×	● (II)
olaparib, pembrolizumab	×	×	● (II)
pamiparib, tislelizumab	×	×	● (II)
sacituzumab govitecan, berzosertib	×	×	● (I/II)
copanlisib, olaparib, durvalumab	×	×	● (I)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Clinical Trials Summary

RAD51D R188Q + RAD51D R53W

NCT ID	Title	Phase
NCT04826341	A Phase I/II Study of Sacituzumab Govitecan Plus Berzosertib in Small Cell Lung Cancer and Homologous Recombination-Deficient Cancers Resistant to PARP Inhibitors	I/II

BRAF V600K

NCT ID	Title	Phase
NCT03898908	Phase II, Multicentre Clinical Trial to Evaluate the Activity of Encorafenib and Binimetinib Administered Before Local Treatment in Patients With BRAF Mutant Melanoma Metastatic to the Brain.	II
NCT02231775	Neoadjuvant and Adjuvant Dabrafenib and Trametinib in Patients With Clinical Stage III Melanoma (Combi-Neo)	II
NCT05263453	A Single-arm, Multi-center Phase II Trial to Evaluate the Efficacy and Safety of the Combination of HL-085 and Vemurafenib in Advanced Melanoma Patients With BRAF V600E/K Mutation	II
NCT02968303	Phase II Study With COmbination of Vemurafenib With Cobimetinib in B-RAF V600E/K Mutated Melanoma Patients to Normalize LDH and Optimize Nivolumab and Ipilimumab therapy	II
NCT04655157	A Multi-Center Phase I/II Open Label Study to Evaluate Safety and Efficacy in Participants With Metastatic BRAF-mutant Melanoma Treated With Encorafenib With and Without Binimetinib in Combination With Nivolumab and Low-dose Ipilimumab. (QUAD 01: Quadruple Therapy in Melanoma)	I/II
NCT04557956	Phase I/II Study of an EZH2 Inhibitor (Tazemetostat) in Combination With Dual BRAF/MEK Inhibition in Patients With BRAF- Mutated Metastatic Melanoma Who Progressed on Prior BRAF/MEK Inhibitor Therapy	I/II
No NCT ID	Phase I Study of Safety and Immune Effects of an Escalating Dose of Autologous GD2 Chimeric Antigen Receptor-Expressing Peripheral Blood T cells in Patients with Metastatic Melanoma	I
NCT04722575	A Phase II Randomized Non-Comparative Study, With Neoadjuvant Plus Adjuvant Therapy With Combination Or Sequence Of Vemurafenib, Cobimetinib, And Atezolizumab In Patients With High-Risk, Surgically Resectable BRAF Mutated And Wild-Type Melanoma	II
NCT05026983	Phase II Study of Binimetinib With Encorafenib in Patients With Metastatic Melanoma and CNS Metastases	II
NCT05299580	"Dabrafenib and Trametinib in Circulating Free DNA BRAFV600 Mutated Metastatic Melanoma Patients: a Prospective Phase II, Open Label, Multicentre Study - (Bioliquid TAILOR Study - BIOTAILOR)"	II
NCT04221438	A Phase II Neoadjuvant Study of Encorafenib With Binimetinib in Patients With Resectable Locoregional Metastases From Cutaneous or Unknown Primary Melanoma (Stages III N1B/C/D)	II
NCT05114603	An Open-label, Multicenter Phase II Clinical Study to Evaluate Safety, Efficacy and PK of HLX208 for Advanced Melanoma With BRAF V600 Mutation	II
NCT04720768	A Phase IB Study to Evaluate the Safety, Tolerability and Preliminary Efficacy of the Combination of Encorafenib, Binimetinib and Palbociclib in Patients With BRAF-mutant Metastatic Melanoma (The CELEBRATE Study)	I/II
NCT03374839	Combined Therapy of Nivolumab and Adoptive T Cell Therapy in Metastatic Melanoma Patients: Pilot Study Phase I/II	I/II

Clinical Trials Summary (continued)

BRAF V600K (continued)

NCT ID	Title	Phase
NCT04190628	A Phase I, First-In-Human, Multicenter, Open-Label Study of ABM-1310, Administered Orally in Adult Patients With Advanced Solid Tumors	I
NCT04249843	A First-in-Human, Phase Ia/Ib, Open Label, Dose-Escalation and Expansion Study to Investigate the Safety, Pharmacokinetics, and Antitumor Activity of the RAF Dimer Inhibitor BGB-3245 in Patients With Advanced or Refractory Tumor	I
NCT04741997	A Randomized Pilot Trial of Adjuvant Therapy Based on Pathologic Response After Neoadjuvant Encorafenib and Binimetinib in Advanced Melanoma	I
NCT03284502	A Phase Ib, Open-label, Multicenter, Dose Escalation Study of the Safety, Tolerability, and Pharmacokinetics of HM95573 in Combination With Either Cobimetinib or Cetuximab in Patients With Locally Advanced or Metastatic Solid Tumors	I
NCT04418167	A Phase I Study of ERK1/2 Inhibitor JSI-1187 Administered as Monotherapy and in Combination With Dabrafenib for the Treatment of Advanced Solid Tumors With MAPK Pathway Mutations	I
NCT04903119	A Phase I Study of Nilotinib in Combination With Dabrafenib and Trametinib in BRAF V600 Mutant Metastatic Melanoma After Progression on BRAF/MEK Inhibition	I
NCT04543188	A Two-Part, Phase Ia/b, Open-Label, Multicenter Trial Evaluating Pharmacokinetics, Safety and Efficacy of PF 07284890 (ARRY 461) in Participants with Braf V600 Mutant Solid Tumors with and without Brain Involvement	I
NCT03554083	Vemurafenib, Cobimetinib, Atezolizumab, and Tiragolumab in Treating Patients With High-Risk Stage III Melanoma	II
NCT04940299	A Phase II Study to Assess the Safety and Efficacy of Tocilizumab in Combination With Ipilimumab and Nivolumab in Patients With Advanced Melanoma, Non-Small Cell Lung Cancer, or Urothelial Carcinoma	II
NCT05270044	Adjuvant Encorafenib & Binimetinib vs. Placebo in Fully Resected Stage IIB/C BRAF V600E/K Mutated Melanoma: a Randomized Triple-blind Phase III Study in Collaboration With the EORTC Melanoma	III
NCT04657991	A Phase III, Randomized, Double-blind Study of Encorafenib and Binimetinib Plus Pembrolizumab Versus Placebo Plus Pembrolizumab in Participants With BRAF V600E/K Mutation-Positive Metastatic OR Unresectable Locally Advanced Melanoma	III
NCT04901988	Circulating Tumour DNA Guided Therapy For Stage IIB/C Melanoma After Surgical Resection (DETECTION)	II/III
NCT04551521	Continuous ReAssessment With Flexible ExTension in Rare Malignancies - CRAFT: The NCT-PMO-1602 Phase II Trial	II
NCT02693535	Targeted Agent and Profiling Utilization Registry (TAPUR) Study	II
NCT03235245	Combination of Targeted Therapy (Encorafenib and Binimetinib) Followed by Combination of Immunotherapy (Ipilimumab and Nivolumab) vs Immediate Combination of Immunotherapy in Patients With Unresectable or Metastatic Melanoma With BRAF V600 Mutation : an EORTC Randomized Phase II Study (EBIN)	II

Clinical Trials Summary (continued)

BRAF V600K (continued)

NCT ID	Title	Phase
No NCT ID	A Phase IA/B Open-label Study to Evaluate Safety, Pharmacokinetics, and Preliminary Clinical Activity of R07276389 alone and in Combination with Cobimetinib in Participants with BRAF-V600 Mutation-positive Advanced Solid Tumor or BRAF-V600 Mutation-positive Melanoma with Central Nervous System Metastases	I
NCT03297606	Canadian Profiling and Targeted Agent Utilization Trial (CAPTUR): A Phase II Basket Trial	II
NCT04511013	A Randomized Phase II Trial of Encorafenib + Binimetinib + Nivolumab vs Ipilimumab + Nivolumab in BRAF-V600 Mutant Melanoma With Brain Metastases	II
NCT04116541	MegaMOST - A Multicenter, Open-label, Biology Driven, Phase II Study Evaluating the Activity of Anti-cancer Treatments Targeting Tumor Molecular Alterations /Characteristics in Advanced / Metastatic Tumors.	II
No NCT ID	Single Arm, Open Label, Phase II Trial Of Vemurafenib And Cobimetinib In Patients With Metastatic Non-Squamous Non-Small Cell Lung Cancer And Other Tumours Harboursing BRAF V600 Mutations Detected By Comprehensive Genomic Profiling	II
NCT04965220	A Phase Ib/II Clinical Trial Evaluating the Safety, Tolerability, Pharmacokinetics, and Initial Efficacy of HLX208 (BRAF V600E Inhibitor) in Combination With Trametinib in Patients With Advanced Solid Tumors	I/II
NCT03781219	A Phase I, Single Arm, Dose Escalation Study to Evaluate Safety, Pharmacokinetics and Preliminary Efficacy of HL-085 Plus Vemurafenib in Patients With BRAF V600 Mutant Advanced Solid Tumor	I
NCT05355701	A PHASE I OPEN-LABEL, DOSE ESCALATION AND DOSE EXPANSION STUDY TO EVALUATE THE SAFETY, TOLERABILITY, PHARMACOKINETICS, AND ANTI TUMOR ACTIVITY OF PF-07799933 (ARRY-440) AS A SINGLE AGENT AND IN COMBINATION THERAPY IN PARTICIPANTS 16 YEARS AND OLDER WITH ADVANCED SOLID TUMORS WITH BRAF ALTERATIONS	I
NCT05340621	A Phase Ib/II Study of OKI-179 Plus Binimetinib in Patients With Advanced Solid Tumors and Activating Mutations in the RAS Pathway (Phase 1b) and in Patients With Advanced NRAS-Mutated Melanoma (Phase II)	I/II
NCT05111561	A Phase I Study of ZEN003694 in Combination With Binimetinib in Solid Tumors With RAS Pathway Alterations and Triple Negative Breast Cancer	I
NCT04423185	Platform Study of Genotyping Guided Precision Medicine for Rare Tumors in China	II
NCT04913285	A Phase 1/1b Open-label, Multicenter Study to Investigate the Safety, Tolerability, Pharmacokinetics, and Antitumor Activity of KIN-2787 in Participants With BRAF and/or NRAS Mutation-positive Solid Tumors.	I

RAD51D R188Q, RAD51D R53W

NCT ID	Title	Phase
NCT04992013	Genomically Guided Phase II Study to Evaluate the Clinical Benefit of Niraparib in Tumors Metastatic to the CNS	II

Clinical Trials Summary (continued)

RAD51D R188Q, RAD51D R53W (continued)

NCT ID	Title	Phase
NCT04985721	An Open Label, Signal Seeking, Translational, Phase II Trial of Pamiparib in Combination With Tislelizumab in Patients With Advanced Tumours With Homologous Recombination Repair Defects	II
NCT03842228	A Phase Ib Biomarker-Driven Combination Trial of Copanlisib, Olaparib, and Durvalumab (MEDI4736) in Patients With Advanced Solid Tumors	I
NCT04423185	Platform Study of Genotyping Guided Precision Medicine for Rare Tumors in China	II
NCT03297606	Canadian Profiling and Targeted Agent Utilization Trial (CAPTUR): A Phase II Basket Trial	II
NCT04123366	A Phase II Study of Olaparib in Combination With Pembrolizumab in Participants With Previously Treated, Homologous Recombination Repair Mutation (HRRm) and/or Homologous Recombination Deficiency (HRD)-Positive Advanced Cancer	II

Assay Information

Methodology, Scope, and Application: The OCAv3 next-generation sequencing (NGS) assay identifies more than 3000 annotated mutations of interest (MOIs) broadly categorized into 5 mutation types: single nucleotide variants (SNVs), small insertions/deletions (Indels), large (>3 bases) insertions/deletions (Large Indels), copy number variants (CNVs), and gene fusions. The assay utilizes the Thermo Fisher Scientific Oncomine® Comprehensive Assay v3.0 (OCAv3), a next-generation sequencing (NGS) assay that utilizes a multiplex polymerase chain reaction (PCR) with DNA and RNA extracted from formalin-fixed tissue for sequencing on the Ion S5 XL platform and analyzed by the current version of Torrent Suite Software and Ion Reporter. The OCAv3 NGS Assay currently can reliably identify the presence or absence of >3000 known mutations and polymorphisms in the 161 unique genes compared to the Human Reference Genome hg19, including the genes listed below. The OCAv3 assay is a laboratory developed test designed to find gene mutations within tumors (somatic mutations).

Analytical Sensitivity and Specificity: The OCAv3 NGS Assay has been determined to be suitably analytically sensitive and specific for the various types of abnormalities within its reportable range, demonstrating 97.18% sensitivity compared with orthogonal assay results and 100% specificity for the 5 mutation types reported. 99.99% or greater reproducibility has been demonstrated. All quality measures for this assay were within defined assay parameters.

Hereditary and Germline Mutations: This assay examines tumor tissue only and does not examine normal (non-tumor) tissue. Mutations detected by the assay may be present only in the tumor, or in every cell of the body (including non-tumor cells). This test also cannot tell whether a potential germline mutation causes or will cause a hereditary cancer syndrome. If the patient's history includes any one or combination of features suggestive of a possible hereditary cancer predisposition (such as, cancer arising at a young age, especially a cancer of a type unusual in younger patients; a personal history of multiple different types of cancers; or a significant cancer history among blood relatives, especially but not exclusively of the same type as the patient's) it is recommended that the physician arrange for the patient to meet with a genetic counselor and, if warranted, undergo the appropriate genetic test on normal (i.e. non-tumor) tissue (blood or cells brushed from the oral surface of the cheek) to check for a germline abnormality, regardless of the results of this research study.

Report Content: Thermo Fisher Scientific's Ion Torrent Oncomine Reporter software was used in the generation of this report. Software was developed and designed internally by Thermo Fisher Scientific. The analysis was based on the current version of Oncomine Reporter. The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. FDA information was sourced from www.fda.gov and is current as of this version. NCCN information was sourced from www.nccn.org and reflects its current version. Clinical Trials information reflects the current version. For the most up-to-date information regarding a particular trials, search www.clinicaltrials.gov by NCT ID or search local clinical trials authority website by local identifier listed in 'Other Identifiers.' Variants are reported according to HGVS nomenclature and classified following AMP/ASCO/CAP guidelines (Li et al.

2017). Based on the data sources selected, variants, therapies, and trials identified in this report are listed in order of potential clinical significance but not for predicted efficacy of the therapies.

Genes assayed for hotspot mutations:

AKT1, AKT2, AKT3, ALK, AR, ARAF, AXL, BRAF, BTK, CBL, CCND1, CDK4, CDK6, CHEK2, CSF1R, CTNNB1, DDR2, EGFR, ERBB2, ERBB3, ERBB4, ERCC2, ESR1, EZH2, FGFR1, FGFR2, FGFR3, FGFR4, FLT3, FOXL2, GATA2, GNA11, GNAQ, GNAS, H3F3A, HIST1H3B, HNF1A, HRAS, IDH1, IDH2, JAK1, JAK2, JAK3, KDR, KIT, KNSTRN, KRAS, MAGOH, MAP2K1, MAP2K2, MAP2K4, MAPK1, MAX, MDM4, MED12, MET, MTOR, MYC, MYCN, MYD88, NFE2L2, NRAS, NTRK1, NTRK2, NTRK3, PDGFRA, PDGFRB, PIK3CA, PIK3CB, PPP2R1A, PTPN11, RAC1, RAF1, RET, RHEB, RHOA, ROS1, SF3B1, SMAD4, SMO, SPOP, SRC, STAT3, TERT, TOP1, U2AF1, XPO1

Genes assayed with full exon coverage:

ARID1A, ATM, ATR, ATRX, BAP1, BRCA1, BRCA2, CDK12, CDKN1B, CDKN2A, CDKN2B, CHEK1, CREBBP, FANCA, FANCD2, FANCI, FBXW7, MLH1, MRE11, MSH2, MSH6, NBN, NF1, NF2, NOTCH1, NOTCH2, NOTCH3, PALB2, PIK3R1, PMS2, POLE, PTCH1, PTEN, RAD50, RAD51, RAD51B, RAD51C, RAD51D, RB1, RNF43, SETD2, SLX4, SMARCA4, SMARCB1, STK11, TP53, TSC1, TSC2

Genes assayed for copy number variations:

AKT1, AKT2, AKT3, ALK, AR, AXL, BRAF, CCND1, CCND2, CCND3, CCNE1, CDK2, CDK4, CDK6, EGFR, ERBB2, ESR1, FGF19, FGF3, FGFR1, FGFR2, FGFR3, FGFR4, FLT3, IGF1R, KIT, KRAS, MDM2, MDM4, MET, MYC, MYCL, MYCN, NTRK1, NTRK2, NTRK3, PDGFRA, PDGFRB, PIK3CA, PIK3CB, PPARG, RICTOR, TERT

Genes assayed for detection of fusions:

AKT2, ALK, AR, AXL, BRAF, BRCA1, BRCA2, CDKN2A, EGFR, ERBB2, ERBB4, ERG, ESR1, ETV1, ETV4, ETV5, FGFR1, FGFR2, FGFR3, FGR, FLT3, JAK2, KRAS, MDM4, MET, MYB, MYBL1, NF1, NOTCH1, NOTCH4, NRG1, NTRK1, NTRK2, NTRK3, NUTM1, PDGFRA, PDGFRB, PIK3CA, PPARG, PRKACA, PRKACB, PTEN, RAD51B, RAF1, RB1, RELA, RET, ROS1, RSPO2, RSPO3, TERT

DISCLAIMER

This assay is considered a Laboratory Developed Test (LDT). Its performance characteristics have been determined through extensive testing although it has not been cleared by the US Food and Drug Administration and such approval is not required for clinical implementation. Furthermore, any comments included in the report are strictly interpretive and the opinion of the reviewer. This laboratory is certified under the Clinical Laboratory Improvements Amendments (CLIA) for the performance of high-complexity testing for clinical purposes. The correlative data presented is a result of the curation of published data sources, but may not be exhaustive. The content of this report has not been evaluated or approved by the FDA or other regulatory agencies.

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